

Java Script Topics – Level-wise Complete Roadmap

Below is a **complete, level-wise roadmap of JavaScript**, from absolute beginner to advanced & professional level. This can be used for **learning, revision, interviews, or roadmap planning**.

● LEVEL 1: Beginner (JavaScript Fundamentals)

Basics

- What is JavaScript?
- JavaScript History & Versions (ES5, ES6+)
- How JavaScript Works in Browser
- JavaScript Engines
- `script` tag (internal, external)
- Comments

Variables & Data Types

- `var`, `let`, `const`
- Naming conventions
- Primitive Data Types
 - Number
 - String
 - Boolean
 - Undefined
 - Null
 - Symbol
 - BigInt
- Non-Primitive Data Types
 - Object
 - Array

Operators

- Arithmetic Operators
- Assignment Operators
- Comparison Operators
- Logical Operators
- Unary Operators
- Ternary Operator

Control Flow

- `if`, `else`, `else if`

Java Script Topics – Level-wise Complete Roadmap

- switch
- Loops
 - for
 - while
 - do...while
 - for...in
 - for...of
- break, continue

Type Conversion

- Implicit Type Coercion
- Explicit Type Conversion

Basic Functions

- Function Declaration
- Function Expression
- Arrow Functions (Basic)
- Parameters & Arguments
- Default Parameters
- Return Statement

Loops (All Types)

- for loop
- while loop
- do...while loop
- for...in loop
- for...of loop
- Nested Loops
- Loop Control
 - break
 - continue

🌀 LEVEL 2: Intermediate JavaScript

Arrays

- Creating Arrays
- Iterating Arrays
 - for
 - forEach

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- map
 - Array Methods
 - push, pop, shift, unshift
 - map, filter, reduce
 - find, findIndex
 - some, every
 - includes, indexOf
 - slice, splice
 - Array Destructuring
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● LEVEL 3: Advanced JavaScript

Execution Context

- Global Execution Context
- Function Execution Context
- Call Stack

Hoisting

- Variable Hoisting
- Function Hoisting

Closures

- Definition
- Practical Examples
- Use Cases

this Keyword (Advanced)

- Global Context
- Function Context
- Object Context
- Arrow Function Behavior
- call, apply, bind

Prototypes & Inheritance

- Prototype Chain
- Constructor Functions
- Prototypal Inheritance

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ES6+ Features

- Modules (import, export)
- Spread Operator
- Enhanced Object Literals
- Optional Chaining
- Nullish Coalescing

Error Handling

- try...catch
- finally
- Custom Errors

🌀 LEVEL 4: Asynchronous JavaScript

Asynchronous Concepts

- Synchronous vs Asynchronous
- Blocking vs Non-blocking

Callbacks

- Callback Functions
- Callback Hell

Promises (Complete)

- What is a Promise?
- Promise States
 - Pending
 - Fulfilled
 - Rejected
- Creating Promises
- then(), catch(), finally()
- Promise Chaining
- Promise APIs
 - Promise.all()
 - Promise.allSettled()
 - Promise.race()
 - Promise.any()

Async / Await

- async functions

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- await keyword
- Sequential vs Parallel Execution
- Error Handling with Async/Await

Event Loop (Detailed)

- Call Stack
 - Web APIs
 - Callback Queue (Task Queue)
 - Microtask Queue
 - Starvation
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● LEVEL 5: DOM & Browser JavaScript

DOM Basics

- What is DOM?
- DOM Tree

DOM Selection

- getElementById()
- getElementsByClassName()
- getElementsByTagName()
- querySelector()
- querySelectorAll()

DOM Manipulation

- Creating Elements
- Appending & Removing Elements
- innerHTML
- textContent
- classList
- setAttribute()

Events

- Event Listeners
- Event Object
- Event Types
- Event Bubbling
- Event Capturing
- Event Delegation

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Forms & Inputs

- Form Submission
- Input Events
- Client-side Validation

Browser APIs

- Fetch API
 - localStorage
 - sessionStorage
 - Cookies
 - Geolocation API
 - History API
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● LEVEL 6: Object-Oriented JavaScript

OOP Concepts

- Encapsulation
- Abstraction
- Inheritance
- Polymorphism

Classes

- Class Syntax
 - Constructors
 - Methods
 - Getters & Setters
 - Static Methods
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● LEVEL 7: Advanced & Professional Topics

Memory Management

- Garbage Collection
- Memory Leaks

Performance Optimization

- Debouncing
- Throttling

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- Lazy Loading

Design Patterns

- Module Pattern
- Factory Pattern
- Singleton Pattern
- Observer Pattern

JavaScript Security

- XSS
- CSRF
- CORS

Testing

- Unit Testing
- Jest Basics
- Test Driven Development (TDD)

Tooling

- Babel
- Webpack
- Vite
- ESLint
- Prettier

BONUS: JavaScript for Interviews

- Shallow vs Deep Copy
- == vs ===
- null vs undefined
- Currying
- Memoization
- Event Loop Questions
- Output-Based Questions

NEXT STEPS

- JavaScript + HTML + CSS Projects
- Learn a Framework (React, Vue, Angular)

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- Backend JavaScript (Node.js)
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M / SahnaWaz